

CSI247: Data Structures (3 credits)
University of Botswana
2021/22

Course Information

Code	CSI247
Title	Data Structures
Credits	3
Type	Core
Semester in which course is taught (1 or 2)	1
Pre-requisites	CSI142, ISS112
Co-requisites	none

Aim and Learning Objectives

This course will introduce students to fundamental data structures and data abstraction, algorithm analysis and design paradigms.

Learning Outcomes

After completion of the course, students should be able to

1. describe common applications for each data structure discussed
2. implement user-defined data structures in a high-level language
3. write programs using each data structure discussed
4. choose appropriate data structures for any given problem
5. perform worst-case runtime analysis on data structures and simple algorithms

Course Synopsis

Analysis of algorithms (worst-case); Lists; Stacks and queues; Trees; Graphs; Hashing; Priority queues; Applications of data structures.

Course delivery

2 hours Lectures, 2 hour Lab/Tutorial

Modes of Assessment

CA and Examination

Reading (and other resources) list

1. Weiss M., "Data Structures and Problem Solving Using Java", 3rd Edition, Addison Wesley, 2005.
2. Goodrich M. and Tamassia R., "Data Structures and Algorithms in Java", 4th Edition, John Wiley and Sons, Inc., 2006.
3. Cormen T., Leiserson C., Rivest R. and Stein C., "Introduction to Algorithms", 2nd Edition, MIT Press, 2001.
4. E-resources.